LAYERED PERMUTATION COPULAS ARE CLOSER TO THE UNIFORM LAW THAN THE DIAGONAL

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Abstract. Schrott conjectured that among all copulas on the unit square, the diagonal and the antidiagonal couplings are the farthest from the uniform law in the quadratic Wasserstein distance, at squared distance 1/10. By density of permutation copulas the conjecture is implied by a statement about n equally weighted grid points: for every permutation σ of $\{0,\dots,n-1\}$, the points $\big((i+\frac{1}{2})/n,(\sigma(i)+\frac{1}{2})/n\big)$ are no farther from the uniform law than the diagonal points. We prove this, with strict inequality, for every layered permutation, that is, every permutation obtained by cutting $0,\dots,n-1$ into runs of consecutive values and reversing some of the runs, other than the identity and the full reversal. The proof re-cuts the optimal partition of the identity on the union of the moved strips, writes the gain as a weighted sum of two-cell comparisons, and settles each comparison by a dispersive-order inequality between the laws of x_1+x_2 and x_1-x_2 on a diagonal band. The same re-cut, done in two steps inside a window of three or four strips, gives the strict inequality for every permutation that splits into blocks of consecutive values of length at most eight (each block reversed or permuted arbitrarily), once $n\ge 9$; for blocks of length four to eight the positivity of explicit functionals is proved with interval arithmetic.

1. The problem

Let λ be Lebesgue measure on $Q=[0,1]^2$. A copula is a probability measure on Q with uniform marginals. Schrott [1] proved that among all couplings of n standard Gaussians, the ones farthest from the standard Gaussian on $\mathbb{R}^n$ in W_2 are the signed diagonals, and stated the uniform analogue as a conjecture.

Conjecture 1.1 ([1, Conjecture 1.1])**.** *For every copula π on Q, $W_2^2(\pi,\lambda)\le 1/10$, with equality for the diagonal coupling.*

For a probability measure μ on Q write $V(\mu)=\max\mathbb{E}[X\cdot Z]$ over couplings of $X\sim\lambda$ and $Z\sim\mu$. Since $W_2^2(\mu,\lambda)=\mathbb{E}|X|^2+\mathbb{E}|Z|^2-2V(\mu)$ and $\mathbb{E}|Z|^2=2/3$ for every copula, Conjecture 1.1 says that $V(\pi)\ge V(\text{diagonal})$ for every copula π.

Fix $n\ge 2$, put $a_i=(i+\frac{1}{2})/n$ for $0\le i\le n-1$, and for a permutation σ of $\{0,\dots,n-1\}$ let $\mu_\sigma=\frac{1}{n}\sum_i\delta_{p_i}$ with $p_i=(a_i,a_{\sigma(i)})$, and $A_n(\sigma)=V(\mu_\sigma)$. Both marginals of μ_σ are the uniform law on the grid $\{a_i\}$, so $\mathbb{E}|Z|^2$ does not depend on σ, and $A_n(\sigma)\ge A_n(\mathrm{id})$ says that μ_σ is no farther from λ than the diagonal points. Consider

$$(\mathrm{A}_n)\qquad A_n(\sigma)\ge A_n(\mathrm{id})\qquad\text{for every permutation }\sigma\text{ of }\{0,\dots,n-1\}.$$

Every copula π is a weak limit of measures μ_{σ_n}. Indeed, let $Z^1,\dots,Z^n$ be independent with law π, almost surely with distinct coordinates, and let σ_n send the rank of Z_1^j to the rank of Z_2^j (ranks counted from 0). Replacing each coordinate by $(\mathrm{rank}+\frac{1}{2})/n$ moves it by at most $\sup_t|F_n(t)-t|+\frac{1}{2n}$, where F_n is the empirical distribution function of that coordinate, which tends to 0 almost surely by the Glivenko–Cantelli theorem; and the empirical measure of the Z^j tends weakly to π. On Q weak convergence is W_2 convergence, and $|V(\mu)-V(\mu')|\le(\mathbb{E}|X|^2)^{1/2}W_2(\mu,\mu')$. Hence, if

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(A_n) holds for all n, then $V(\pi) = \lim_n A_n(\sigma_n) \geq \lim_n A_n(\mathrm{id}) = V(\text{diagonal}) = 37/60$ for every copula π, the last limit by the same argument applied to the diagonal. So (A_n) for all n implies Conjecture 1.1.

The map A_n is invariant under the eight symmetries of the square acting on permutation matrices, so $A_n(\mathrm{rev}) = A_n(\mathrm{id})$ for the full reversal $\mathrm{rev}(i) = n-1-i$. Numerically (A_n) holds for every permutation when $n \leq 8$, with equality only at id and rev (Section 5).

Definition 1.2. A permutation σ is *layered* if there are integers $0 = k_0 < k_1 < \cdots < k_m = n$ such that, for each ℓ, σ maps $\{k_\ell, \ldots, k_{\ell+1} - 1\}$ onto itself either identically or by reversal.

Theorem 1.3. *Let $n \geq 2$ and let σ be a layered permutation of $\{0, \ldots, n-1\}$ other than* id *and* rev. *Then $A_n(\sigma) > A_n(\mathrm{id})$. The same holds for the images of σ under the symmetries of the square.*

For $n = 2$ every permutation is id or rev, so the theorem is about $n \geq 3$. Section 4 extends Theorem 1.3 to permutations built from blocks of length at most eight (Theorem 4.9).

2. The identity and the re-cut

Write $s = x_1 + x_2$ and $v = x_1 - x_2$ for $x \in Q$. Under λ, $S = X_1 + X_2$ has density $L(s) = \min(s, 2-s)$ on $[0,2]$, and given $S = s$ the variable $V = X_1 - X_2$ is uniform on $[-L(s), L(s)]$. Let $0 = q_0 < q_1 < \cdots < q_n = 2$ be the quantiles of S at levels j/n and $T_j = \{x \in Q : q_j < s < q_{j+1}\}$.

Lemma 2.1. $A_n(\mathrm{id}) = \sum_j a_j \int_{T_j} s\,dx$.

Proof. A coupling of λ with μ_σ gives each point p_i a sub-probability $\gamma_i \leq \lambda$ of mass $1/n$, and $\mathbb{E}[X \cdot Z] = \sum_i \int x \cdot p_i \, d\gamma_i$. For $\sigma = \mathrm{id}$, $x \cdot p_j = a_j s$, and with $a_0 < \cdots < a_{n-1}$ the sum $\sum_j a_j \int s\, d\gamma_j$ under $\sum_j \gamma_j = \lambda$, $\gamma_j(Q) = 1/n$, is largest when γ_j is λ on T_j (the rearrangement inequality applied to the law of S). □

Any partition of Q into regions $R_0, \ldots, R_{n-1}$ of area $1/n$ gives the coupling $\gamma_i = \lambda|_{R_i}$, so $A_n(\sigma) \geq \sum_i \int_{R_i} x \cdot p_i \, dx$.

Consider a block of $r \geq 2$ consecutive values $k, \ldots, k+r-1$ with $k + r \leq n$, and let σ reverse it and fix every other value. Its moved points are $p_{k+i} = (a_{k+i}, a_{k+r-1-i})$, $0 \leq i \leq r-1$; they share $p_1 + p_2 = 2\bar{a}$ with $\bar{a} = a_k + (r-1)/(2n)$, and $p_1 - p_2 = w_i := (2i+1-r)/n$, increasing in i. Keep T_j for every fixed point, and cut $U = T_k \cup \cdots \cup T_{k+r-1} = \{q_k < s < q_{k+r}\}$ into the regions $R_0, \ldots, R_{r-1}$ of area $1/n$ cut out by consecutive quantiles of v on U, from the smallest values of v up; give R_i to p_{k+i}. Since $x \cdot p = \frac{1}{2}\big(s(p_1+p_2) + v(p_1 - p_2)\big)$, Lemma 2.1 gives $A_n(\sigma) - A_n(\mathrm{id}) \geq D$ with

$$D = \sum_{i=0}^{r-1} \int_{R_i} \Big(\bar{a}s + \tfrac{1}{2} w_i v\Big) dx - \sum_{j=k}^{k+r-1} a_j \int_{T_j} s\, dx = \frac{1}{n^2}\Big(\sum_{i=0}^{r-1} (i - \tfrac{r-1}{2})\, \nu_i - \sum_{i=0}^{r-1} (i - \tfrac{r-1}{2})\, m_i\Big),$$

where $m_i = n \int_{T_{k+i}} s\, dx$ is the mean of S on the strip T_{k+i} and $\nu_i = n \int_{R_i} v\, dx$ the mean of V on R_i.

Lemma 2.2. *For real numbers $c_0, \ldots, c_{r-1}$, $\sum_i (i - \frac{r-1}{2}) c_i = \sum_{t=1}^{r-1} \frac{t(r-t)}{r}\, \Delta_t(c)$ with $\Delta_t(c) = \frac{1}{r-t} \sum_{i \geq t} c_i - \frac{1}{t} \sum_{i < t} c_i$.*

Proof. With $\bar{c}$ the mean, $\sum_i i(c_i - \bar{c}) = \sum_{t \geq 1} \sum_{i \geq t} (c_i - \bar{c})$, and $\sum_{i \geq t} (c_i - \bar{c}) = (r-t)\big(\frac{1}{r-t} \sum_{i \geq t} c_i - \bar{c}\big) = \frac{t(r-t)}{r} \Delta_t(c)$. □

So $n^2 D = \sum_{t=1}^{r-1} \frac{t(r-t)}{r}(\Delta_t(\nu) - \Delta_t(m))$. Both ν and m are the means of a variable over its own quantile cells of mass $1/n$ on U: $\Delta_t(\nu)$ is the gap between the mean of V above and below its quantile at mass t/n on U, and $\Delta_t(m)$ the same gap for S.

3. A dispersive comparison on diagonal bands

Fix $0 \le \alpha < \beta \le 2$ with $(\alpha, \beta) \ne (0, 2)$ and let $U = \{\alpha < s < \beta\}$, of area ρ. Let Φ_S and Φ_V be the distribution functions of S and V under $\lambda|_U$ (total mass ρ), Q_S, Q_V their inverses on $(0, \rho)$, and $\phi_S = L \circ Q_S$, $\phi_V = g \circ Q_V$ the densities read at mass level μ, where $g(y) = \frac{1}{2}|(\alpha, \beta) \cap (|y|, 2 - |y|)|$ is the density of V under $\lambda|_U$. Write $\mu' = \rho - \mu$.

Lemma 3.1. *For $0 < \mu < \rho$, $\phi_V(\mu) \le \sqrt{2\min(\mu, \mu')} \le \phi_S(\mu)$. If $\alpha > 0$ there is $c > 0$ with $\phi_V(\mu) < \phi_S(\mu)$ for $0 < \mu < c$, and if $\beta < 2$ there is $c > 0$ with $\phi_V(\mu) < \phi_S(\mu)$ for $\rho - c < \mu < \rho$.*

Proof. Let $s = Q_S(\mu)$. If $s \le 1$, then $\mu = \int_\alpha^s L \le \int_0^s t\, dt = s^2/2$, so $\phi_S(\mu) = s \ge \sqrt{2\mu}$, strictly if $\alpha > 0$. If $s \ge 1$, then $\mu' = \int_s^\beta L \le (2 - s)^2/2$ and $\phi_S(\mu) = 2 - s \ge \sqrt{2\mu'}$, strictly if $\beta < 2$. In both cases $\phi_S(\mu) \ge \sqrt{2\min(\mu, \mu')}$.

For the left inequality, g is even. For $y \in [-1, 0]$ the interval $(-y, 2 + y)$ grows with y and each of its ends moves at unit speed, so g is nondecreasing on $[-1, 0]$, $g(-1) = 0$, and $g(y) - g(z) \le y - z$ for $-1 \le z \le y \le 0$. Hence for $y = Q_V(\mu) \le 0$,

$$\mu = \int_{-1}^{y} g(z)\, dz \ge \int_{-1}^{y} \big(g(y) - (y - z)\big)_+ dz \ge \tfrac{1}{2} g(y)^2,$$

so $\phi_V(\mu) \le \sqrt{2\mu}$, and $\mu \le \mu'$ by the symmetry of g. The case $y \ge 0$ is the mirror image.

For the strict inequality let $\alpha > 0$. If $\alpha < 1$, put $c = \min(\Phi_S(1), \rho) > 0$: for $\mu < c$ we have $Q_S(\mu) < 1$ and $\phi_S(\mu) > \sqrt{2\mu} \ge \phi_V(\mu)$ by the first paragraph. If $\alpha \ge 1$, put $c = \rho/2$: then $Q_S(\mu) \ge 1$, and for $\mu < c$ we have $\mu < \mu'$, so $\phi_S(\mu) \ge \sqrt{2\mu'} > \sqrt{2\mu} \ge \phi_V(\mu)$. The case $\beta < 2$ follows by applying the map $(x_1, x_2) \mapsto (1 - x_2, 1 - x_1)$, which preserves λ, fixes v, sends s to $2 - s$ and the band (α, β) to $(2 - \beta, 2 - \alpha)$, and reverses the mass levels of S. □

Proposition 3.2. *Let $n \ge 3$, $2 \le r \le n - 1$, $0 \le k \le n - r$. Then $D > 0$ for the reversal of the block $k, \dots, k + r - 1$.*

Proof. Take $\alpha = q_k$, $\beta = q_{k+r}$; since $r < n$, $(\alpha, \beta) \ne (0, 2)$. Since $Q_S' = 1/\phi_S$ and $Q_V' = 1/\phi_V$, Lemma 3.1 shows that $h = Q_V - Q_S$ is nondecreasing on $(0, \rho)$. For $\tau = t/n$,

$$\Delta_t(\nu) - \Delta_t(m) = \frac{1}{\tau(\rho - \tau)} \int_0^\tau \int_\tau^\rho \big(h(\mu) - h(\mu_0)\big)\, d\mu\, d\mu_0 \ge 0.$$

If $\alpha > 0$, Lemma 3.1 gives $c > 0$ with h strictly increasing on $(0, c)$, so $h(\mu) > h(\mu_0)$ whenever $\mu_0 < \min(\tau, c)/2$ and $\mu > \tau$; the integral is then positive for every t. If $\alpha = 0$ then $\beta < 2$, h is strictly increasing on some $(\rho - c, \rho)$, and $h(\mu) > h(\mu_0)$ whenever $\mu_0 < \tau$ and $\mu > \rho - \min(\rho - \tau, c)/2$. By Lemma 2.2, $n^2 D > 0$. □

Proof of Theorem 1.3. Let the reversed runs of σ be $B_1, \dots, B_m$, of sizes $r_\ell \ge 2$. If some $r_\ell = n$ then $\sigma = \mathrm{rev}$; otherwise each $r_\ell \le n - 1$, and $m \ge 1$ since $\sigma \ne \mathrm{id}$. The unions U_ℓ of the strips of the runs are disjoint, so performing the re-cut of Section 2 inside every U_ℓ and keeping the other strips is a partition of Q into regions of area $1/n$ matched to the points of μ_σ. Its value is $A_n(\mathrm{id}) + \sum_\ell D_\ell$, and each $D_\ell > 0$ by Proposition 3.2. The symmetries of the square preserve λ and the grid, so A_n is invariant under them. □

Remark 3.3. For $r = 2$ at the corner ($k = 0$ or $k = n - 2$, $n \ge 4$) the bound is exactly $n^2 D = \frac{2\sqrt{2} - 2}{3}\, n^{-1/2} \approx 0.276\, n^{-1/2}$, while the measured gap $n^2(A_n(\sigma) - A_n(\mathrm{id}))$ is about $0.344\, n^{-1/2}$ ($n = 8, 12, 16$). The corner transposition is the cheapest move, so the margin in (A_n) near the identity is of order $n^{-5/2}$.

4. Windows of three to six values

Let $r \geq 2$, let τ be a permutation of $\{0, \dots, r-1\}$, $0 \leq k \leq n-r$, and let $\sigma(k+i) = k + \tau(i)$ for $0 \leq i \leq r-1$, σ the identity elsewhere. Put $u_i' = i + \tau(i) - (r-1)$ and $v_i' = i - \tau(i)$. The moved points satisfy $p_1 + p_2 = 2\bar{a} + u_i'/n$ and $p_1 - p_2 = v_i'/n$ for $p = p_{k+i}$, with $\bar{a}$ and $U = \{q_k < s < q_{k+r}\}$ as in Section 2. For a region C of positive area write $\mathbb{E}_C$ for the average over C with respect to λ.

Lemma 4.1. *Let $C_0, \dots, C_{r-1}$ be a partition of U into regions of area $1/n$, and*

$$F = \sum_{i=0}^{r-1} (u_i' \, \mathbb{E}_{C_i}[S] + v_i' \, \mathbb{E}_{C_i}[V]) - \sum_{j=0}^{r-1} (2j - r + 1) \, \mathbb{E}_{T_{k+j}}[S].$$

Then $A_n(\sigma) - A_n(\mathrm{id}) \geq F/(2n^2)$. If σ acts in this way on several disjoint blocks, re-cutting each block's strips separately gives $A_n(\sigma) - A_n(\mathrm{id}) \geq \sum_\ell F_\ell/(2n^2)$.

Proof. Give C_i to p_{k+i} and keep T_j for the fixed points. With $x \cdot p = \frac{1}{2}(s(p_1 + p_2) + v(p_1 - p_2))$ and $\int_C g \, dx = \frac{1}{n}\mathbb{E}_C[g]$, $\sum_i \int_{C_i} x \cdot p_{k+i} \, dx = \bar{a} \int_U s \, dx + \frac{1}{2n^2} \sum_i (u_i' \mathbb{E}_{C_i}[S] + v_i' \mathbb{E}_{C_i}[V])$, while by Lemma 2.1 the identity collects $\sum_j a_{k+j} \int_{T_{k+j}} s \, dx = \bar{a} \int_U s \, dx + \frac{1}{2n^2} \sum_j (2j - r + 1) \mathbb{E}_{T_{k+j}}[S]$. Disjoint blocks use disjoint unions of strips, so the gains add. □

For the reversal of the block, $u_i' = 0$ and $v_i' = 2i - r + 1$, and $F/(2n^2)$ is the quantity D of Section 2. The strips T_{k+j} are the cells of U between consecutive quantiles of S. We use two partitions. *v then s*: group the indices i by the value of v_i'; cut U by V at its quantiles of mass $1/n$ times the cumulative group sizes, the lowest values of V going to the group with the smallest v'; cut each group's region by S into cells of area $1/n$ and give them to the group's indices in increasing order of u_i'. *s then v* is the same with the roles of (S, u') and (V, v') exchanged. The band U is symmetric under $v \mapsto -v$, so the median of V on U is 0.

The symmetries $(x_1, x_2) \mapsto (x_2, x_1)$ and $(x_1, x_2) \mapsto (1 - x_2, 1 - x_1)$ preserve λ, map strips to strips and windows to windows, and replace τ by τ^{-1} and by $i \mapsto r - 1 - \tau^{-1}(r - 1 - i)$; their composite gives $i \mapsto r - 1 - \tau(r - 1 - i)$. The second sends the band $\{\alpha < s < \beta\}$ to $\{2 - \beta < s < 2 - \alpha\}$. The image of a partition is a partition with the same value of F. So it suffices to treat one block in each orbit, on every band.

Proposition 4.2. *Let $n \geq 4$ and $\tau = (1, 2, 0)$ or $(2, 0, 1)$. Then the v then s partition gives $F > 0$ on every band U of three consecutive strips.*

Proof. The two 3-cycles form one orbit; take $\tau = (1, 2, 0)$, with $(u_i', v_i') = (-1, -1), (1, -1), (0, 2)$. Let c be the upper tercile of V on U (so $c \geq 0$), $C_2 = U \cap \{v > c\}$, and split the rest $U' = U \setminus C_2$ at the median of S on U' into C_0 (lower) and C_1 (upper). Since $\mathbb{E}_U[V] = 0$, $\mathbb{E}_{C_0}[V] + \mathbb{E}_{C_1}[V] = 2\mathbb{E}_{U'}[V] = -\mathbb{E}_{C_2}[V]$, and

$$F = 3 \, \mathbb{E}_{C_2}[V] + (\mathbb{E}_{C_1}[S] - \mathbb{E}_{C_0}[S]) - 2(\mathbb{E}_{T_{k+2}}[S] - \mathbb{E}_{T_k}[S]) = \tfrac{3}{2}(G_V - G_S) + (\Delta_{U'}(S) - \tfrac{1}{2} G_S),$$

where G_X is the mean of X on the upper third of U (by mass) minus its mean on the lower third, and $\Delta_{U'}(S)$ the gap between the means of S above and below its median on U'. With $h = Q_V - Q_S$ nondecreasing (Lemma 3.1), $G_V - G_S = \frac{3}{\rho}(\int_{2\rho/3}^{\rho} h - \int_0^{\rho/3} h) \geq 0$. For the second bracket, the section of U' over s has length $L(s) + \min(L(s), c) \geq L(s)$, half the section of U. For any law, the gap between the means above and below the median is $2 \min_t \mathbb{E}|X - t|$; the mass of U' is $2/3$ that of U, so $\Delta_{U'}(S) \geq \frac{3}{4}\Delta_U(S)$. The upper half of U is its upper third together with the upper half of its middle third, in proportions $2 : 1$, and likewise below, so $\Delta_U(S) > \frac{2}{3} G_S$. Hence $\Delta_{U'}(S) - \frac{1}{2} G_S > 0$ and $F > 0$. □

For $r = 4$ there are thirteen indecomposable permutations (no proper initial segment $\{0, \dots, j-1\}$ is mapped onto itself); one is the reversal and the other twelve fall into six orbits,

$$\{1302, 2031\},\ \{1320, 2130, 3021, 3102\},\ \{3120\},\ \{2301\},\ \{2310, 3201\},\ \{1230, 3012\},$$

writing τ as the word $\tau(0)\tau(1)\tau(2)\tau(3)$. On a band U let $S_1, \dots, S_4$ be the means of S on the four strips, $d_{41} = S_4 - S_1$, $d_{32} = S_3 - S_2$, x_m the median of $|V|$ on U, A and B the means of $|V|$ above and below x_m, T_{out} and T_{in} the means of S on $\{|v| > x_m\}$ and $\{|v| < x_m\}$, and R_1, R_3 the means of S on the lowest and highest thirds (by mass, in s) of $\{v < x_m\}$. By the symmetry $v \mapsto -v$ the quartiles of V on U are $-x_m, 0, x_m$.

Lemma 4.3. *The v then s partition gives*

$$\begin{aligned}
F(1302) &= 4A + 2B - 3d_{41} - d_{32}, & F(1320) &= 5A + B + (T_{\text{out}} - T_{\text{in}}) - 3d_{41} - d_{32},\\
F(2301) &= 4A + 4B - 2d_{41}, & F(2310) &= 5A + 3B - \tfrac{5}{2}d_{41} - \tfrac{1}{2}d_{32},\\
F(1230) &= 4A + 2(R_3 - R_1) - 3d_{41} - d_{32}, & F(3120) &= 6A + (T_{\text{hi}} - T_{\text{lo}}) - 3d_{41} - d_{32},
\end{aligned}$$

where $T_{\text{hi}}, T_{\text{lo}}$ are the means of S above and below its median on $\{|v| < x_m\}$.

Proof. For 1302 the points are $(u', v') = (-2, -1), (1, -2), (-1, 2), (2, 1)$; the v' are distinct, so the cells are the quartile cells of V, $\{v < -x_m\}, \{-x_m < v < 0\}, \{0 < v < x_m\}, \{v > x_m\}$, carrying $v' = -2, -1, 1, 2$ and $u' = 1, -2, 2, -1$. The V-means are $-A, -B, B, A$, which gives $4A + 2B$; the S-means are $T_{\text{out}}, T_{\text{in}}, T_{\text{in}}, T_{\text{out}}$, and the weights $1 - 1$ and $-2 + 2$ cancel them. The strips contribute $-(3S_4 + S_3 - S_2 - 3S_1) = -3d_{41} - d_{32}$. The other five are the same computation. For 1320, $(u', v') = (-2, -1), (1, -2), (1, 0), (0, 3)$, the cells are again the V-quartiles. For 2301 the points are $(\pm 1, \pm 2)$: the halves $\{v < 0\}$, $\{v > 0\}$ are split at the S-median, which is that of U, and the V-means add to $2(A + B)$ on each side. For 2310, $(u', v') = (-1, -2), (1, -2), (0, 1), (0, 3)$, the half $\{v < 0\}$ is split by S and the upper half into $\{0 < v < x_m\}$ and $\{v > x_m\}$. For 1230, $(u', v') = (-2, -1), (0, -1), (2, -1), (0, 3)$, the top quarter is $\{v > x_m\}$ with V-mean A, the rest $\{v < x_m\}$ has V-mean $-A/3$ and is cut into S-thirds. For 3120, $(u', v') = (0, -3), (-1, 0), (1, 0), (0, 3)$, the outer quarters go to $v' = \pm 3$ and $\{|v| < x_m\}$ is split at its S-median. □

Proposition 4.4. *Let τ be one of the twelve indecomposable non-layered permutations of $\{0, 1, 2, 3\}$. On every band U of mass $\rho \le 4/9$, the v then s partition gives $F > 0$.*

Proof. By the orbits it suffices to treat the six representatives. Always $A \ge x_m \ge B \ge 0$, $T_{\text{hi}} > T_{\text{lo}}$, $R_3 > R_1$ and $d_{41} > d_{32} > 0$. Lemma 4.3 gives

$$\begin{aligned}
F(3120) - F(1302) &= 2(A - B) + (T_{\text{hi}} - T_{\text{lo}}), & F(2301) - F(1302) &= 2B + d_{41} + d_{32},\\
F(2310) - F(1302) &= A + B + \tfrac{1}{2}(d_{41} + d_{32}), & F(1230) &\ge F_0 := 4A - 3d_{41} - d_{32},
\end{aligned}$$

all differences positive, and $F(1302) = F_0 + 2B \ge F_0$. So it remains to show $F_0 > 0$ and $F(1320) > 0$ for $\rho \le 4/9$. Write $U = \{\alpha < s < \beta\}$ and $w = \beta - \alpha$.

Thin bands across the diagonal, $\alpha < 1 < \beta$, $w < 1/3$. On U, $L \ge 1 - w$, so $\rho \ge w(1 - w)$. The distribution function of $|V|$ on U is $H(x) = \int_\alpha^\beta \min(L(s), x)\, ds \le wx$, and $H(x_m) = \rho/2$, so $x_m \ge (1 - w)/2$ and $A \ge (1 - w)/2$. All S_j, T and R lie in (α, β), so each difference of them is at most w in absolute value. Hence $F_0 \ge 2(1 - w) - 4w > 0$ and $F(1320) \ge \frac{5}{2}(1 - w) - 5w > 0$.

Bands on one side of the diagonal, $\beta \le 1$ (the case $\alpha \ge 1$ follows by $s \mapsto 2 - s$, which leaves A, B, d_{41}, d_{32} unchanged and reverses the sign of $T_{\text{out}} - T_{\text{in}}$). Every quantity in the functionals is a length, and the band $\{\alpha < s < \beta\}$ is the band $\{\alpha/\beta < s < 1\}$ scaled by β, so each functional equals $\beta G(\alpha/\beta)$ for an explicit algebraic function G on $[0, 1)$: the quantiles of S and of $|V|$ solve quadratic equations, and the means are ratios of polynomial integrals. We enclosed G

on subintervals of $[0,1]$ in interval arithmetic with outward rounding (`mpmath.iv`), bisecting until the lower end of the enclosure is positive: F_0 in 24 boxes (smallest lower bound 0.035), $F(1320)$ in 12 boxes on $\beta \le 1$ (0.124) and 17 boxes on $\alpha \ge 1$ (0.172). Since this covers every $t = \alpha/\beta \in [0,1]$, it holds for bands of any mass.

Wider bands across the diagonal, $\alpha < 1 < \beta$, $\rho \le 4/9$, $w \ge w_0$. In the coordinates $(\alpha, 2-\beta) \in [0,1]^2$, $\rho = (2-\alpha^2-(2-\beta)^2)/2$, and ρF is a combination of the integrals $K(x) = \int_0^x sL(s)\,ds$ at α, β and the quartiles of S, of $N(x) = \int_0^x y\,dH(y)$ at x_m and 1, and, for 1320, of $P(x_m) = \int_\alpha^\beta s\,(L(s)-x_m)_+\,ds$, which gives T_{out}. The quartiles of S increase in α and β, K and N increase and P decreases in x, and x_m is enclosed by bisection on the sign of $H(x) - \rho/2$ over a whole box; so each box of parameters gets an enclosure of ρF. Bisecting boxes until the enclosure is positive: F_0 with $w_0 = 0.05$ in 2,774 boxes, $F(1320)$ with $w_0 = 0.25$ in 1,042 boxes, no failures. Together with the thin case this covers every band of mass at most 4/9. □

For $r = 5$ there are 71 indecomposable permutations; one is the reversal, and the other 70 fall into 25 orbits under the symmetries of the square. On a band U let $V_1, \dots, V_5$ be the quintile cells of V, from the lowest values to the highest, A and B the means of V on V_5 and V_4, and s_1, s_2, s_3 the means of S on V_5, V_4 and V_3. By the symmetry $v \mapsto -v$ the means of V on V_1, V_2 are $-A, -B$ and the means of S on V_1, V_2 are s_1, s_2. The *quintile partition* gives $V_1, \dots, V_5$ to the indices $i_1, \dots, i_5$ listed in increasing order of v'_i, tied indices in any order. By Lemma 4.1 it gives

$$F_q = (v'_{i_5} - v'_{i_1})A + (v'_{i_4} - v'_{i_2})B + (u'_{i_1} + u'_{i_5})s_1 + (u'_{i_2} + u'_{i_4})s_2 + u'_{i_3}s_3 - \sum_{j=0}^{4}(2j-4)S_j,$$

with $S_0, \dots, S_4$ the means of S on the five strips. For $\tau = 12403$ the points are $(u', v') = (-3,-1), (-1,-1), (2,-2), (-1,3), (3,1)$, and the s then v partition gives the three outer strips to the indices $0, 2, 4$ and splits the two middle strips $T_{k+1} \cup T_{k+2}$ at $v = 0$, the lower half to index 1 and the upper half to index 3. Its value is

$$F_s = 4\,\mathbb{E}\big[\,|V| \,\big|\, S \in (x_1, x_3)\big] - (S_2 - S_1) - (S_4 - S_0),$$

where $x_1 < x_2 < x_3 < x_4$ are the quintiles of S on U.

Proposition 4.5. *Let τ be one of the 70 indecomposable permutations of $\{0, \dots, 4\}$ other than the reversal. On every band U of mass $\rho \le 5/9$ some partition gives $F > 0$: the s then v partition for the orbit $\{12403, 14023, 20341, 30142\}$, and for each of the other 24 orbits a quintile partition.*

Proof. By the orbits it suffices to treat one representative of each orbit. Write $U = \{\alpha < s < \beta\}$ and $w = \beta - \alpha$.

The orbit of 12403. Given $S = s$, $|V|$ is uniform on $(0, L(s))$, so $\mathbb{E}[|V| \mid S \in (x_1, x_3)]$ is a ratio of $\int L^2/2$ and $\int L$ over (x_1, x_3). On a band with $\beta \le 1$ every quantity is a length, $F_s = \beta\, G(\alpha/\beta)$ with $x_p = (t^2 + p(1-t^2))^{1/2}$ for the band $\{t < s < 1\}$, and the means are $(x^2+xy+y^2)/(3(x+y))$ and $2(x^2+xy+y^2)/(3(x+y))$ on an interval (x, y), so G is a closed form with no singularity on $[0,1]$. For $\alpha \ge 1$, $s \mapsto 2 - s$ maps the band to a band below the diagonal, leaves V unchanged and sends strip j to strip $4 - j$, which gives a second closed form. Both were enclosed in interval arithmetic on subintervals of $[0,1]$: 8 boxes below the diagonal (smallest lower bound 0.018) and 7 above it (0.221), so these hold for bands of any mass. Across the diagonal, with $\Lambda(x) = \int_0^x L^2/2$ and $K(x) = \int_0^x sL(s)\,ds$,

$$\tfrac{\rho}{5}F_s = 2\Lambda(x_3) - 2\Lambda(x_1) - K(x_3) + 2K(x_2) + K(x_4) - K(\alpha) - K(\beta),$$

which decreases in x_1 and x_3 (the derivatives are $-L(x_1)^2$ and $L(x_3)(L(x_3) - x_3)$) and increases in x_2 and x_4. The quintiles increase in α and β, so each box of $(\alpha, 2-\beta)$ gets a lower bound from the ends of the quintile enclosures. With $w_0 = 0.05$, 415 boxes, no failures. When $w < w_0$,

U lies in $(1-w_0, 1+w_0)$, so $L \geq 1-w_0$ and $4\mathbb{E}[|V| \mid \cdot] \geq 2(1-w_0)$, while each difference of means of S is less than w_0; hence $F_s \geq 2-4w_0 > 0$.

The other orbits. The coefficients satisfy $v'_{i_5} - v'_{i_1} > 0$, $v'_{i_4} - v'_{i_2} \geq 0$, and the coefficients of s_1, s_2, s_3 sum to $\sum_i u'_i = 0$. On bands on one side of the diagonal $F_q = \beta\, G(\alpha/\beta)$ as before; the quintiles of V on the band $\{t < s < 1\}$ have two closed forms each, with the branch changing at $t = 1/9$ and $t = 3/7$, and G was enclosed on the three pieces of $[0,1]$ in both orientations for all 24 representatives: 1,960 boxes, smallest lower bound 0.0034. Across the diagonal, $\frac{\rho}{5}F_q$ is a combination of K at α, β and the quintiles of S, and of the integrals of $|V|$ and of S over $\{|V| < y\}$ at $y = 1$ and at the points $y_1 > y_2$ where the distribution function H of $|V|$ equals $\frac{3}{5}\rho$ and $\frac{1}{5}\rho$. Its derivative in y_1 is $\frac{1}{2}h(y_1)((c_B - c_A)y_1 + (d_2 - d_1)\bar{s}(y_1))$ and in y_2 is $\frac{1}{2}h(y_2)(-c_B y_2 + (2d_3 - d_2)\bar{s}(y_2))$, where c_A, c_B are the coefficients of A, B, d_1, d_2, d_3 those of s_1, s_2, s_3, $h = H'$ and $\bar{s}(y)$ the mean of S on $\{|V| = y\}$; where the bracket keeps one sign over a box the corresponding end of the enclosure of y_1 or y_2 is used, and otherwise the whole enclosure. With $w_0 = 0.05$ the 24 representatives take 112,654 boxes, no failures (smallest lower bound of F_q: $4.9 \cdot 10^{-6}$). When $w < w_0$: $\rho \geq w(1-w)$, the distribution function of $|V|$ is at most wx, so y_1, the upper quintile of V, is at least $\frac{3}{5}(1-w)$ and $A \geq \frac{3}{5}(1-w)$, $B \geq 0$; every mean of S lies in (α, β), so the s-terms are at least $-Uw$ with U the sum of the positive u'_i, and the strip term lies in $[0, 6w]$. Hence $F_q \geq \frac{3}{5}(v'_{i_5} - v'_{i_1})(1-w) - (U+6)w$, which decreases in w and is positive at $w = 0.05$ in all 24 cases (smallest value 1.73). □

Proposition 4.6. *Let τ be one of the* 460 *indecomposable permutations of $\{0, \dots, 5\}$ other than the reversal. On every band U of mass $\rho \leq 6/9$ the following partition gives $F > 0$ for a suitable assignment of its cells to the indices: cut U at the terciles of S and each of the three pieces at $V = 0$.*

Proof. Write $U = \{\alpha < s < \beta\}$, let $\alpha = x_0 < x_1 < \cdots < x_6 = \beta$ be the sextiles of S on U, and let e_j and ℓ_j be the means of S and of $L(S)$ on the strip (x_j, x_{j+1}). The six cells have mass $\rho/6 = 1/n$. Given $S = s$, V is uniform on $(-L(s), L(s))$, so the cell of the k-th third on the side $\pm V > 0$ has $\mathbb{E}[S] = \frac{1}{2}(e_{2k} + e_{2k+1})$ and $\mathbb{E}[V] = \pm\frac{1}{4}(\ell_{2k} + \ell_{2k+1})$. If these two cells go to the indices i_k^- and i_k^+, then

$$F = \sum_{j=0}^{5}(A_j e_j + B_j \ell_j), \qquad A_j = \tfrac{1}{2}(u'_{i_k^-} + u'_{i_k^+}) - (2j-5), \quad B_j = \tfrac{1}{4}(v'_{i_k^+} - v'_{i_k^-}), \quad k = \lfloor j/2 \rfloor.$$

On an interval $(x, y) \subset (0,1)$ both means equal $2(x^2 + xy + y^2)/(3(x+y))$; on $(x, y) \subset (1,2)$ the mean of S is 2 minus the same expression in $2-y, 2-x$, and the mean of $L(S)$ is 2 minus the mean of S; across 1 both are mixtures of the two pieces, weighted by mass. For $\beta \leq 1$ the similarity $(s, v) \mapsto (s, v)/\beta$ gives $F = \beta\, G(\alpha/\beta)$ with G the value on $\{t < s < 1\}$, $t \in [0,1]$; for $\alpha \geq 1$ the map $s \mapsto 2 - s$ leaves V unchanged and reduces to the same family with every u'_i replaced by $-u'_i$. Across the diagonal the band is given by $(\alpha, 2-\beta) \in [0,1]^2$, and $\rho \leq 6/9$ is $\alpha^2 + (2-\beta)^2 \geq 2/3$. The sextiles increase in α and in β, so a box of parameters gives an enclosure of each x_j from its corners. On each box we take the assignment that is best at the centre, enclose every mean in interval arithmetic and intersect the enclosure with the a priori range of that mean ($e_j \in (x_j, x_{j+1})$, and ℓ_j between the minimum and maximum of L on that strip); the intersection is what controls thin bands across the diagonal, where all the means are close to 1 and F is close to $\frac{1}{2}\sum_k (v'_{i_k^+} - v'_{i_k^-})$. Boxes are bisected until the lower bound is positive. For all 460 blocks this takes 31,694 boxes, with no failures; the smallest lower bound of F is 0.0015. □

Proposition 4.7. *Let τ be one of the* 3,446 *indecomposable permutations of $\{0, \dots, 6\}$ other than the reversal. On every band U of mass $\rho \leq 7/9$ the following partition gives $F > 0$ for a*

suitable assignment of its cells to the indices: cut U at the septiles x_2, x_4, x_6 of S, so that the first three pieces have mass $2\rho/7$ and the last has mass $\rho/7$, and cut each of the first three pieces at $V = 0$.

Proof. With x_j, e_j and ℓ_j now the septiles of S on U and the means on (x_j, x_{j+1}), the seven cells have mass $\rho/7 = 1/n$. As in Proposition 4.6, the cells of the k-th piece ($k = 0, 1, 2$) have $\mathbb{E}[S] = \frac{1}{2}(e_{2k} + e_{2k+1})$ and $\mathbb{E}[V] = \pm\frac{1}{4}(\ell_{2k} + \ell_{2k+1})$, and the last cell has $\mathbb{E}[S] = e_6$ and $\mathbb{E}[V] = 0$. If the cells of the k-th piece go to $i_k^{\pm}$ and the last cell to i_3, then $F = \sum_{j=0}^{6}(A_j e_j + B_j \ell_j)$ with A_j and B_j as in Proposition 4.6 for $j \le 5$ (with $2j - 6$ in place of $2j - 5$), $A_6 = u'_{i_3} - 6$ and $B_6 = 0$. The families of bands are those of Proposition 4.6, with $\rho \le 7/9$ across the diagonal, that is $\alpha^2 + (2 - \beta)^2 \ge 4/9$. The map $s \mapsto 2 - s$ reverses the order of the pieces, so for $\alpha \ge 1$ the reduced family carries the partition with the single piece first, again with every u'_i replaced by $-u'_i$. The enclosures and the bisection are as before. For all 3,446 blocks this takes 315,613 boxes, with no failures; the smallest lower bound of F is 1.2×10^{-4}. □

Proposition 4.8. *Let τ be one of the* 29,092 *indecomposable permutations of $\{0, \dots, 7\}$ other than the reversal. On every band U of mass $\rho \le 8/9$ the following partition gives $F > 0$ for a suitable assignment of its cells to the indices: cut U at the quartiles of S and each of the four pieces at $V = 0$.*

Proof. With x_j, e_j and ℓ_j now the octiles of S on U and the means on (x_j, x_{j+1}), the eight cells have mass $\rho/8 = 1/n$, the cells of the k-th piece ($k = 0, \dots, 3$) have $\mathbb{E}[S] = \frac{1}{2}(e_{2k} + e_{2k+1})$ and $\mathbb{E}[V] = \pm\frac{1}{4}(\ell_{2k} + \ell_{2k+1})$, and $F = \sum_{j=0}^{7}(A_j e_j + B_j \ell_j)$ with A_j and B_j as in Proposition 4.6, with $2j - 7$ in place of $2j - 5$. Across the diagonal $\rho \le 8/9$ is $\alpha^2 + (2 - \beta)^2 \ge 2/9$, and for $\alpha \ge 1$ the map $s \mapsto 2 - s$ carries the partition to itself, with every u'_i replaced by $-u'_i$. Two symmetries reduce the number of blocks. Replacing τ by τ^{-1} replaces each point (u'_i, v'_i) by $(u'_i, -v'_i)$; exchanging the two cells of every piece then gives the same F on every band. Replacing τ by $i \mapsto 7 - \tau(7 - i)$ replaces the points by $(-u'_i, -v'_i)$, which together with the first symmetry is the reduction above: the value on U becomes the value on the reflected band. A proof for τ on all three families is therefore a proof for every block in its orbit under the group of order at most four generated by these maps, and the 29,092 blocks fall into 7,578 orbits. With the enclosures and the bisection as before, one block from each orbit takes 788,143 boxes, with no failures; the smallest lower bound of F is 3.8×10^{-5}. □

Theorem 4.9. *Let $n \ge 9$, and let $\sigma \ne \mathrm{id}, \mathrm{rev}$ map each interval $\{k_\ell, \dots, k_{\ell+1} - 1\}$ of a cut $0 = k_0 < \cdots < k_m = n$ onto itself, acting on each interval either by reversal or by a permutation of an interval of length at most eight. Then $A_n(\sigma) > A_n(\mathrm{id})$, and the same holds for the images of σ under the symmetries of the square. If every interval of length four to eight is reversed or fixed, $n \ge 4$ suffices.*

Proof. Refine the cut so that σ is indecomposable on each interval. An indecomposable piece of length at most eight is a fixed point, a reversal, one of the two 3-cycles, one of the twelve blocks of Proposition 4.4, or one of the seventy blocks of Proposition 4.5, one of the 460 blocks of Proposition 4.6, one of the 3,446 blocks of Proposition 4.7, or one of the 29,092 blocks of Proposition 4.8; a reversed piece of length n would make $\sigma = \mathrm{rev}$. By Lemma 4.1 the gain is at least the sum over the moved pieces of $F_\ell/(2n^2)$, and each F_ℓ is positive: reversals by Proposition 3.2, 3-cycles by Proposition 4.2, the twelve blocks by Proposition 4.4, since four strips have mass $4/n \le 4/9$, and the seventy blocks by Proposition 4.5, since five strips have mass $5/n \le 5/9$, the 460 blocks of six by Proposition 4.6, since six strips have mass $6/n \le 6/9$, the 3,446 blocks of seven by Proposition 4.7, since seven strips have mass $7/n \le 7/9$, and the 29,092 blocks of eight by Proposition 4.8, since eight strips have mass $8/n \le 8/9$. □

For $n \le 8$ every permutation was checked numerically (Section 5). The blocks 1302 and 1320 need the restriction on the mass: their v then s value is negative on nearly full bands, $\{0 < s < 1.8\}$ for 1302 and $\{0.2 < s < 2\}$ for 1320.

5. Computations

All computations use exact moments of S and one-dimensional quadrature of the law of V on a band.

(1) Check of (A_n) over every permutation for $n \le 8$ (one per class under the symmetries of the square; 5,282 classes at $n = 8$), with A_n computed by a smoothed semi-discrete dual on a 40×40 grid of cells. No permutation falls below the identity; the smallest positive gap is the corner adjacent transposition ($n^2(A_n(\sigma) - A_n(\mathrm{id})) = 0.1215$ at $n = 8$), and the first non-layered permutations are corner 3-cycles (0.3944 at $n = 8$).
(2) The bound D for every block reversal, $3 \le n \le 48$, $2 \le r \le n-1$, every k: all positive, with minimum at the corner adjacent transposition, $n^{5/2}D = 0.2761$, matching the closed form above. Quadrature error below 10^{-8}.
(3) The identity of Lemma 2.2 applied to D, checked against the direct evaluation for $n = 5, 8, 13$ and every (k, r): difference below 1.1×10^{-8}; every term $\Delta_t(\nu) - \Delta_t(m)$ is positive.
(4) The closed forms of Lemma 4.3 against a direct evaluation of the v then s cut on polygons, on seven bands: agreement to 10^{-7}; the values agree across each orbit to 5×10^{-15}.
(5) The interval enclosures of Proposition 4.4, in `mpmath.iv`; the scripts are in the repository. As an independent check, $F(1302)$ was also proved positive directly: 17 boxes on one-sided bands (smallest lower bound 0.129) and 1,550 boxes on crossing bands of mass at most 4/9.
(6) For the 70 blocks of five values, the better of the v then s and s then v cuts evaluated on an 11×11 grid of bands of mass at most 5/9: positive for all 70, smallest value of F/ρ equal to 0.756. On the same grid the quintile partition is positive for 24 of the 25 orbits and negative for the orbit of 12403, which is why that orbit takes the s then v partition. For 12403 the closed forms in the proof of Proposition 4.5 agree with the direct evaluation at three bands.
(7) For the 460 blocks of six values, the partition of Proposition 4.6 with the best assignment, on the same kind of grid of bands of mass at most 6/9 (with two thinner masses added): positive for all 460, smallest value of F/ρ equal to 1.04, attained by 123450 and 501234 on $\{0 < s < 2 - \sqrt{2/3}\}$. For the identity the same partition gives negative values, as it must. The closed forms used in the interval enclosures agree with the direct evaluation through the integrals of $sL(s)$ and $L(s)^2$ to 10^{-6} at the bands checked.
(8) For the 3,446 blocks of seven values, the partition of Proposition 4.7 with the best assignment, on a 9×9 grid of bands of mass at most 7/9 with one thin mass added: positive for all 3,446, smallest value of F/ρ equal to 1.04, attained by 1234560 and 6012345 on $\{0 < s < 4/3\}$; the identity is negative. The interval enclosures at point bands contain the values computed from the cell integrals, which agree with the septile means to 10^{-13}, also through the reflection used for $\alpha \ge 1$.
(9) For the 29,092 blocks of eight values, the partition of Proposition 4.8 with the best assignment, on the same grid of bands of mass at most 8/9: positive for all 29,092, smallest value of F/ρ equal to 0.72, attained by 12345670 and 70123456; the identity is negative. The interval enclosures at point bands contain the values computed from the cell integrals, to 4×10^{-13}. The value of F agrees across each orbit of Proposition 4.8 to 6×10^{-12} on 2,914 pairs of a block and a band.

6. What remains

Theorems 1.3 and 4.9 cover the permutations that split into layers and blocks of at most eight values. For blocks of nine or more values the partitions of Propositions 4.6 to 4.8 have natural analogues (pieces of S made of one or two quantile strips, the pairs split at $V = 0$), but their positivity on every band of the allowed mass is open. On the whole square a cut along one direction is not enough: cutting Q into quantile cells of $s\cos\theta + v\sin\theta$ and matching the cells optimally never beats the strips for any θ when $n = 4$. A proof of (A_n) along these lines would need cuts whose direction varies across the square. For $n \leq 8$ the inequality rests on the computation of Section 5.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, found the reduction, the re-cut construction and the proofs, wrote and ran the computations of Section 5, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code for the computations of Sections 4 and 5, including the interval-arithmetic proofs, with its output, is available at `https://github.com/yspennstate/wasserstein-layered-permutation-copulas`.

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